THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS FULL SYLLABUS PAPER SET 01

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The point $(-6, -2)$ lies: 1

(a) Quadrant I **(b)** Quadrant II **(c)** Quadrant IV **(d)** Quadrant III

2. The value of the linear polynomial $3x + 1$ at $x = 4$ is: 1

(a) 16 **(b)** 11 **(c)** 13 **(d)** -13

3. The lines $y = 3x + 2$ and $y = 3x - 5$ are: 1

(a) parallel **(b)** intersecting at the origin

(c) coincident **(d)** perpendicular

4. The absolute value $|-\frac{3}{5}|$ is: 1

(a) $-\frac{3}{5}$ **(b)** $\frac{5}{3}$ **(c)** 0 **(d)** $\frac{3}{5}$

5. $(3s + 5t)(3s - 5t)$ equals: 1

(a) $9s^2 - 25t^2$ **(b)** $9s^2 - 25t^2 + 4$ **(c)** $9s^2 - 30st + 25t^2$ **(d)** $9s^2 + 30st + 25t^2$

6. Through two given points the number of circles that can be drawn is: 1

(a) exactly two **(b)** infinitely many **(c)** exactly one **(d)** none

7. Taking $\pi = \frac{22}{7}$, the area of a circle of radius 14 cm is: 1

(a) 616 cm^2 **(b)** 88 cm^2 **(c)** 308 cm^2 **(d)** 1232 cm^2

8. The number of outcomes in the sample space when two coins are tossed together is: 1

(a) 8 **(b)** 6 **(c)** 4 **(d)** 2

9. The sequence $2, 5, 8, 11, \ldots$ is: 1

(a) an AP with $d = 2$ **(b)** a GP with $r = 3$

(c) neither an AP nor a GP **(d)** an AP with $d = 3$

10. The 25th term of the AP $2, 6, 10, \ldots$ is: 1

(a) 196 **(b)** 97 **(c)** 98 **(d)** 99

11. The proposition "If a number is a multiple of 12, then it is a multiple of 4" is true. Its converse is: 1

(a) false; 8 is a multiple of 4 but not of 12

(b) false; 12 is not a multiple of 4

(c) true; 12 is a multiple of 4

(d) true; every multiple of 4 is a multiple of 12

12. The number 36 has: 1

(a) 9 factors; it is a perfect square **(b)** 11 factors; it is a perfect square
(c) 11 factors; it is not a perfect square **(d)** 10 factors; it is not a perfect square

13. The weighted mean of values $x_1, x_2, \ldots, x_n$ with weights $w_1, w_2, \ldots, w_n$ is: 1

(a) $\dfrac{x_1 + x_2 + \cdots + x_n}{n}$ **(b)** $\dfrac{w_1 + w_2 + \cdots + w_n}{x_1 + x_2 + \cdots + x_n}$ **(c)** $w_1 x_1 \cdot w_2 x_2 \cdots w_n x_n$ **(d)** $\dfrac{w_1 x_1 + w_2 x_2 + \cdots + w_n x_n}{w_1 + w_2 + \cdots + w_n}$

14. 100 units of a mixture with 60% metal is mixed with a mixture of 40% metal to get 50% metal. The quantity of the second mixture needed is (same units): 1

(a) 200 **(b)** 120 **(c)** 100 **(d)** 80

15. A list, which is used to store divisors in order, is an example of a: 1

(a) theorem **(b)** variable **(c)** fraction **(d)** data structure

16. In a parallelogram ABCD, ∠A = (4x + 10)° and ∠B = (2x + 20)°. The value of x is: 1

(a) 15 **(b)** 20 **(c)** 30 **(d)** 25

17. The slope of the line through the points (1, 2) and (3, 8) is: 1

(a) 1/3 **(b)** 4 **(c)** 3 **(d)** -3

18. Which of the following points lies on the line 3x + 2y = 12? 1

(a) (2, 3) **(b)** (2, 2) **(c)** (0, 4) **(d)** (1, 3)

19. The total surface area of a cuboid of length 9 cm, width 7 cm and height 2 cm is: 1

(a) 190 cm^2 **(b)** 95 cm^2 **(c)** 105 cm^2 **(d)** 126 cm^2

20. **Assertion (A):** The surface area of a sphere of radius r equals the curved surface area of a cylinder of radius r and height r. 1

Reason (R): The curved surface area of a cylinder is $2\pi rh$.

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 20–30 words each.

21. The points $A(2,\ 0)$, $B(0,\ 2)$ and $C(-2,\ 0)$ are joined to form a triangle. Name the type of triangle with reason. 2

22. Use Heron's formula to find the area of a triangle of sides 3 cm, 4 cm and 5 cm and verify with $\frac{1}{2}bh$. 2

23. A coin is tossed 200 times and shows heads 112 times. Find the relative frequency of heads and compare it with $\frac{1}{2}$. 2

24. Find the nth term of the AP $11, 8, 5, 2, \ldots$ and write its recursive rule. 2

25. A 100% stacked bar shows Shop P: 60% sales in the morning; Shop Q: 40% in the morning. Can we say P sold more in the morning? Give a reason. 2

26. State Euclid's observation used to reduce gcd(m, n), where m ≥ n. 2

27. In $\triangle ABC$, the line through the midpoint P of AB parallel to BC meets AC at Q. If AC $= 18$ cm, find AQ and QC. 2

28. Solve by adding and subtracting: $x + y = 12$ and $x - y = 4$. 2

SECTION — C (Short Answer Type) 8 × 3 = 24

Answer in about 40–60 words each.

29. A basketball player scored 18 times in 30 attempts. Find the experimental probability of scoring and estimate the number of scores in 50 attempts. 3

30. The sides of $\triangle ABC$ are AB $= 14$ cm, BC $= 18$ cm and CA $= 10$ cm. D, E, F are the midpoints of AB, BC, CA. Find the perimeter of $\triangle DEF$ and the length of each of its sides. 3

31. Show that the average $\frac{a+b}{2}$ of two rational numbers $a < b$ lies between them. 3

32. A state generated 600, 100, 200 and 100 MW from coal, solar, wind and hydro in 2015. Find the percentage shares for a 100% stacked bar. 3

33. Write the coordinates of three points on the line $y = 3x - 2$ and state where it cuts the y-axis. 3

34. The base circumference of a hemispherical dome is 66 m. Find the cost of painting its outer curved surface at ₹40 per m^2 ($\pi = \frac{22}{7}$). 3

35. In a circle of radius 10 cm there are two chords at distances 6 cm and 8 cm from the centre. Find their lengths and show that the nearer one is longer. 3

OR

The angles A and C of a cyclic quadrilateral are in the ratio 4 : 5. Find them. 3

36. **Case study:** Arjun runs Euclid's subtraction algorithm on gcd(7, 3) and Āryabhaṭa's algorithm on the same pair. 3

(i) Write the chain of pairs for subtraction.
(ii) Write the chain for division.
(iii) Which is shorter and by how many reductions?

OR

Use the division algorithm (long division) to find gcd(825, 375) and state the number of reductions. 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 100–150 words each.

37. (i) State the difference in what can be inferred from stacked and 100% stacked bar charts. 5
(ii) Give one question easier to answer with each.

OR

A solution is made by mixing 600 mL of a 5% salt solution with 300 mL of an 8% sugar solution. 5

(i) Find the percentage of salt in the mixture.
(ii) Find the percentage of sugar.
(iii) Why are the answers not 5% and 8%?

38. In quadrilateral ABCD, $AB \parallel DC$ and $AB \neq DC$. G and H are the midpoints of AC and BD. Prove that $GH \parallel AB$ and find GH when AB $= 12$ cm and DC $= 20$ cm. 5

OR

Show how to tile the plane with copies of any given triangle ABC, using parallelograms. 5

39. Show that $0.\overline{142857} = \frac{1}{7}$ and verify the cyclic property by computing 2×142857 and 3×142857. **5**

OR

Case study: Brahmagupta's rules: fortune + fortune = fortune; debt + debt = debt; debt × fortune = debt; debt × debt = fortune. Using these, **5**

(i) compute $(-3) \times 4$ and $(-3) \times (-4)$;

(ii) a spice trader borrows ₹850, earns ₹1,200 and loses ₹450 — find his final standing;

(iii) explain $(-3) \times (-4) = 12$ in terms of debt.

40. The points $(0,\ 5)$ and $(4,\ 13)$ lie on a line $y = ax + b$. **5**

(i) Find a and b.

(ii) Write the equation.

(iii) Is $(10, 25)$ on the line?

(iv) Find the point where the line cuts the x-axis.

OR

Draw the graphs of $y = 3x$ and $y = -3x$ using suitable points. **5**

(i) Where do they meet?

(ii) Compare their slopes.

(iii) State what the sign of the slope tells about growth and decay.

END OF QUESTION PAPER **Total: 80 Marks**

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